QP Code 25901387

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026
Fourth SEMESTER
Discipline Specific Core Course (DSC) - MG4DSCMAT203 - MATHEMATICAL TOOLS FOR COMPUTING
(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

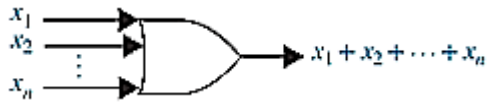
Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A (Short answer type Questions)

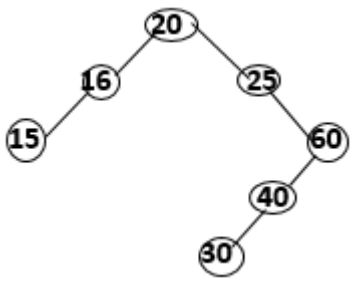
Answer any 5 questions Each question carries 2 marks

1. Lcm $(2^3.3^5.7^2,\ 2^4.3^3.5) = 2^{max(2,4)}.3^{mx)(5,3)}.5^{max(1,0)}.7^{max(2,0)}$
   $= 2^4.3^5.5.7^2$ …….2marks [U] / [CO1]
2. h(45)=45 mod13= 6 ……..1mark
   h(56)= 56 mod 13=4 …… 1mark [A] / [CO1]
3. OR gate with 'n' inputs



……..2marks [A] / [CO2]

4. a) $\overline{(x+0)}$. $(\bar{y}+1)$ …….. 1mark
   b) (x+y) .($\bar{x}$ +$\bar{y}$) ……..1mark [A] / [CO2]
5. R={(1,1),(1,2),(1,3),(1,4),(2,2),(2,4),(3,3),(4,4)} ……2marks [A] / [CO3]
6. R1∪R2 = {(1,1),(2,2),(3,3),(1,2),(1,3),(1,4)} ……1mark
   R1∩R2= {(1,1)} ……..1mark [A] / [CO3]
7. Binary Search Tree (BST)



……..2marks [A] / [CO4]

8. Weighted graph ……. 1mark , Example……..1mark [A] / [CO4]

Part B (Short Essay Type Questions)
Answer any 5 questions Each question carries 6 marks

9. 177mod31=22 and 270mod31=22 ……..2marks

(177mod31·270mod31)mod31 = (22×22)mod31

= 484mod31 = 19 ……4marks [A] / [CO1]

10. To show that gcd(252, 198) = 18 by Euclidean algorithm …….3marks

$$252 = 198 \cdot 1 + 54$$
$$198 = 54 \cdot 3 + 36$$
$$54 = 36 \cdot 1 + 18$$
$$36 = 18 \cdot 2 + 0.$$

Writing $18 = 4 * 252 - 5 * 198$. ……3marks [A] / [CO1]

11. a) $(1.1) + \overline{(1.1) + (0.1)} = 1$ ……3marks

b) $(1.1) + \bar{0} = 1$ ……3marks [A] / [CO2]

12. a) The minterms : xy, $x\bar{y}$, $\bar{x}y$, $\bar{x}\bar{y}$ ……. 3marks

b) F(x,y) = xy + $\bar{x}y$ ……3marks [A] / [CO2]

13. There are loops at a, b, c, and d → (a,a), (b,b), (c,c), (d,d) ∈ R

So the relation R is Reflexive ……1 ½ marks

Between a and c there are arrows both ways → (a,c) and (c,a) ∈ R

Between a and d there are arrows both ways → (a,d) and (d,a) ∈ R

So the relation R is symmetric ……1 ½ marks

(c,a) and (a,d) are in R , But (c,d) is not present

So the relation R is **not Transitive.** ……2marks

So the given relation is not an equivalence relation because it is not transitive

……..1mark [U] / [CO3]

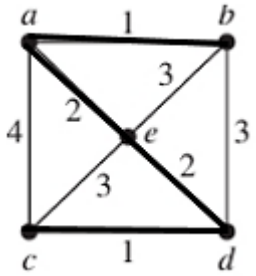
14. Directed edges: 1→2, 1→3, 2→3, 3→2, 3→3 ……3marks

$M_R = \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$ …… 3marks [U] / [CO3]

15. a) Describe the Depth-First Search (DFS) algorithm ……. 2marks

b) Illustrate with a graph ……..4marks [U] / [CO4]

16. Minimum Spanning Tree by Prim's algorithm



Minimum Spanning Tree edges : a−b=1, a−e=2 , e−d=2 , d−c=1 ……4marks

Minimum Spanning Tree Weight **= 6** ……..2marks [U] / [CO4]

Part C (Essay Type Questions)

Answer any 3 questions Each question carries 10 marks

17. First replace the letters in the message with numbers

S	T	O	P		P	O	L	L	U	T	I	O	N
18	19	14	15		15	14	11	11	20	19	8	14	13

……2marks

To replace each of these numbers p by f (p) = (17p + 22) mod 26

P :	18	19	14	15		15	14	11	11	20	19	8	14	13
F(p) :	16	7	0	17		17	0	1	1	24	7	2	0	9

……..5marks

Translating this back to letters produces the encrypted message

STOP POLLUTION encrypts to **Q H A R R A B B Y H C A J** ……..3marks [A] / [CO1]

18. Distributive laws

a) (i) x(y + z) = xy + xz (ii) x+y.z = (x+y) . (x+z) ……..4marks

b) Proof ……6marks [A] / [CO2]

19. $M_{R1UR2} = M_{R1} \vee M_{R2} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$ ……..2marks

$M_{R1nR2} = M_{R1} \wedge M_{R2} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{pmatrix}$ ……..2marks

$M_{R2 \circ R1} = M_{R1} \circ M_{R2} = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{pmatrix}$ ……..2marks*

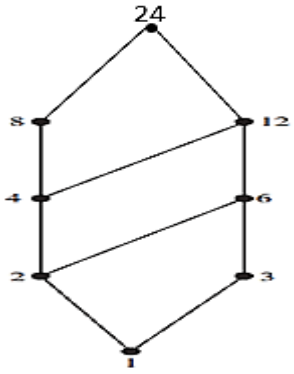
[* Give mark for $M_{R2 \circ R1} = M_{R2} \circ M_{R1} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$

$M_{R1 \circ R1} = M_{R1} \circ M_{R1} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{pmatrix}$ …….2marks

$M_{R1 \oplus R2} = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix}$ …….2marks [A] / [CO3]

20. Define partial ordering. ……2marks

Show that the divisibility relation '|' is a partial ordering on the set of positive integers .
. ……3marks



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Hasse diagram of ({1, 2, 3, 4, 6, 8, 12, 24}, /) …….5marks [A]/[CO3]

21. Briefly explain the preorder, inorder, and postorder traversals of a rooted tree
. ……..4marks

Preorder: **a, b, e, f, l, m, n, c, g, h, o, p, d, i, j, q, s, t, r, k** ……2marks

Inorder: **e, b, l, f, m, n, a, g, c, o, h, p, i, d, s, q, t, j, r, k** ……2marks

Postorder : **e, l, m, n, f, b, g, o, p, h, c, I , s, t, q, r, j, k, d, a** ……2marks
. [A]/[CO4]

22. Explain rooted trees ……2marks

Define and explain parent, child, siblings, ancestors, descendants, internal vertices, leaves, and subtrees with a diagram ……..1mark each (1*8 = 8marks) [U] / [CO4]